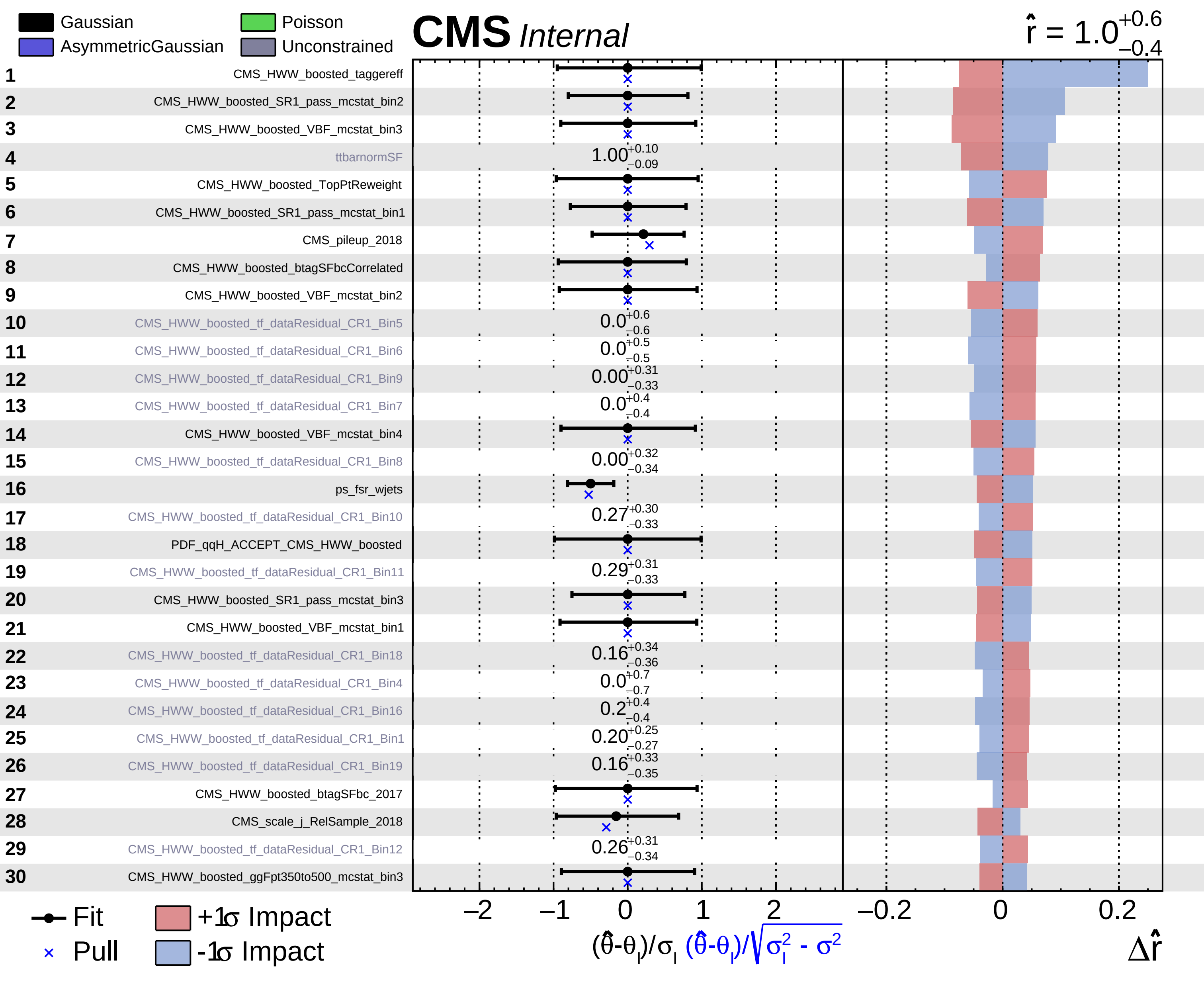
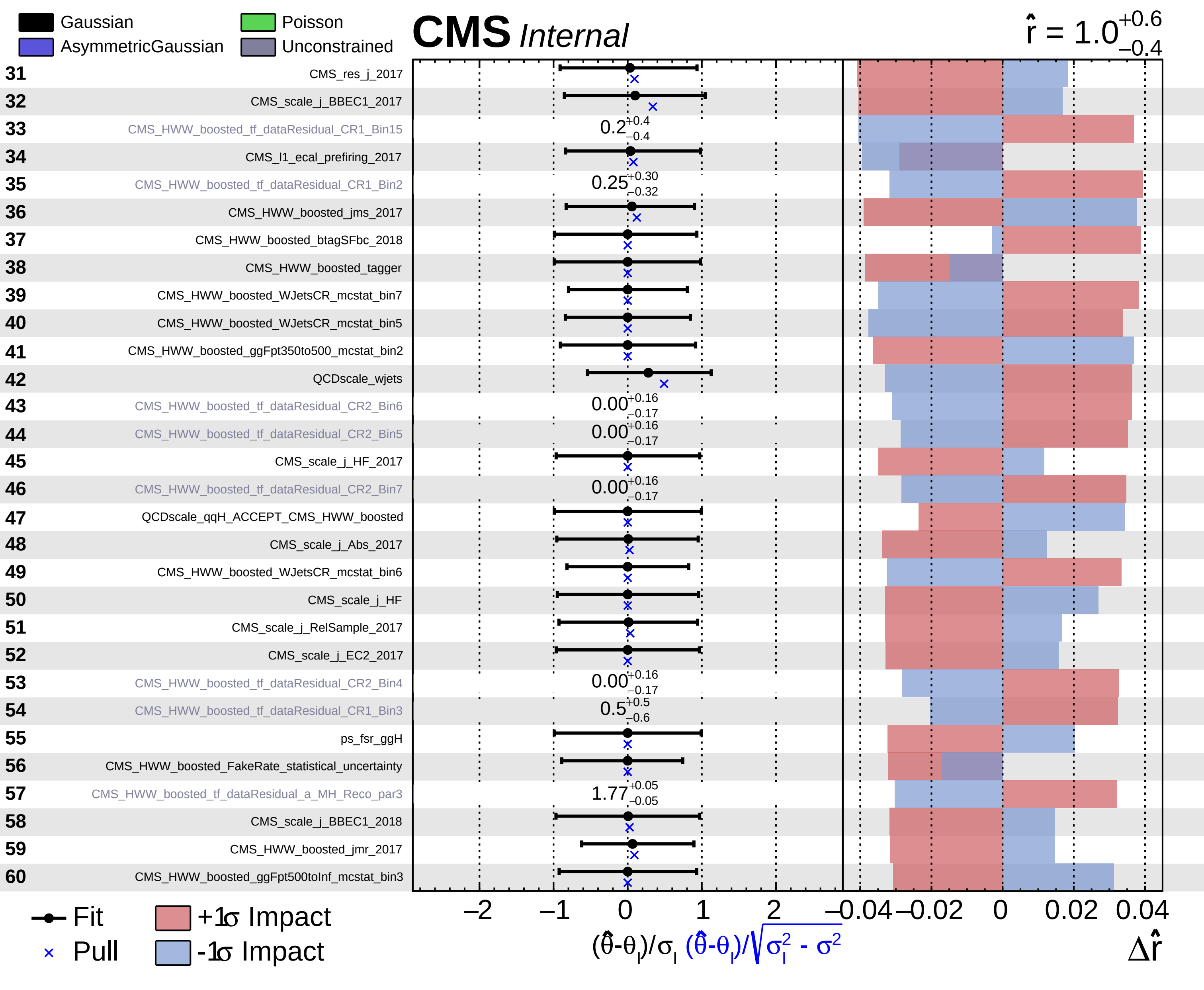
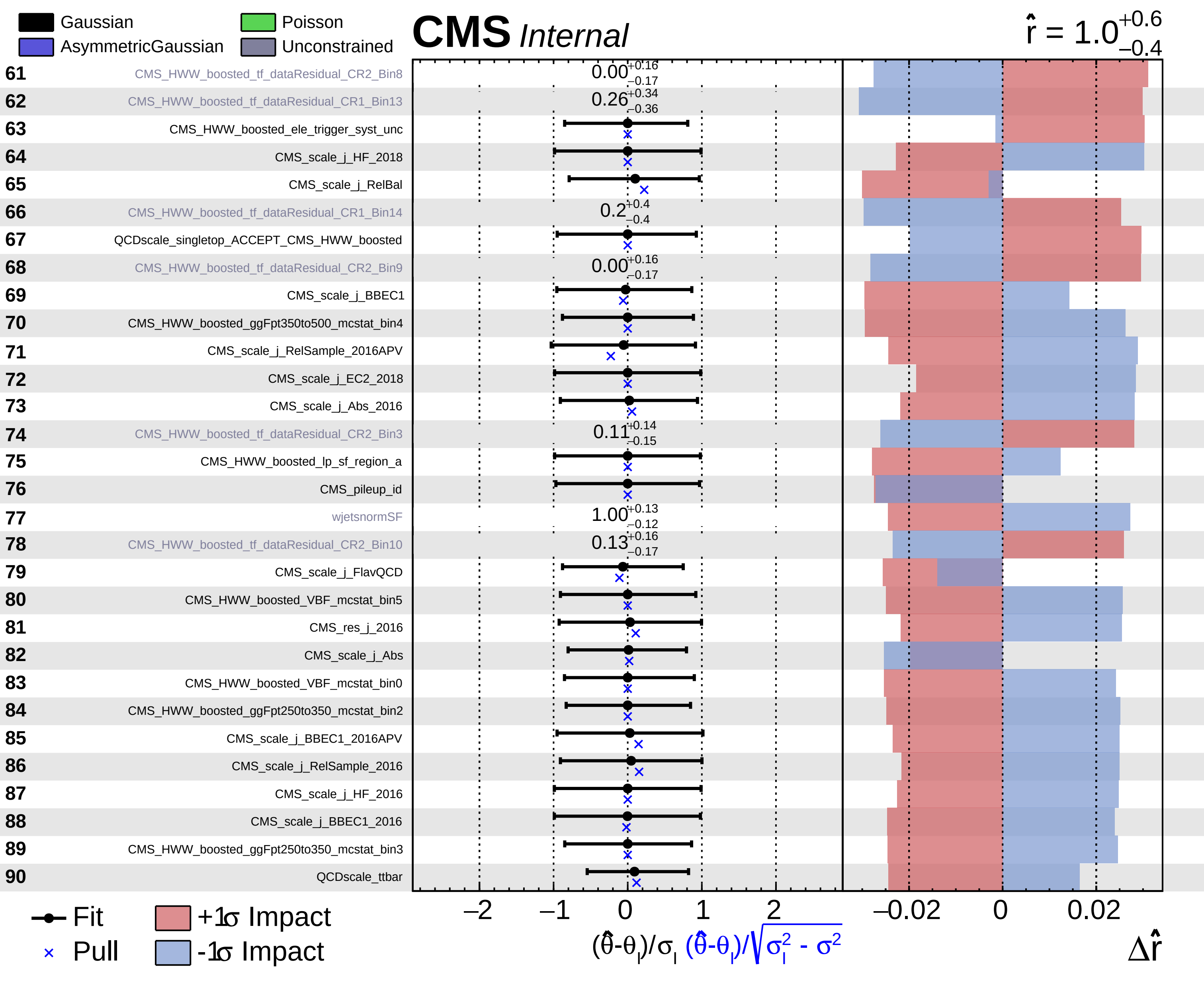
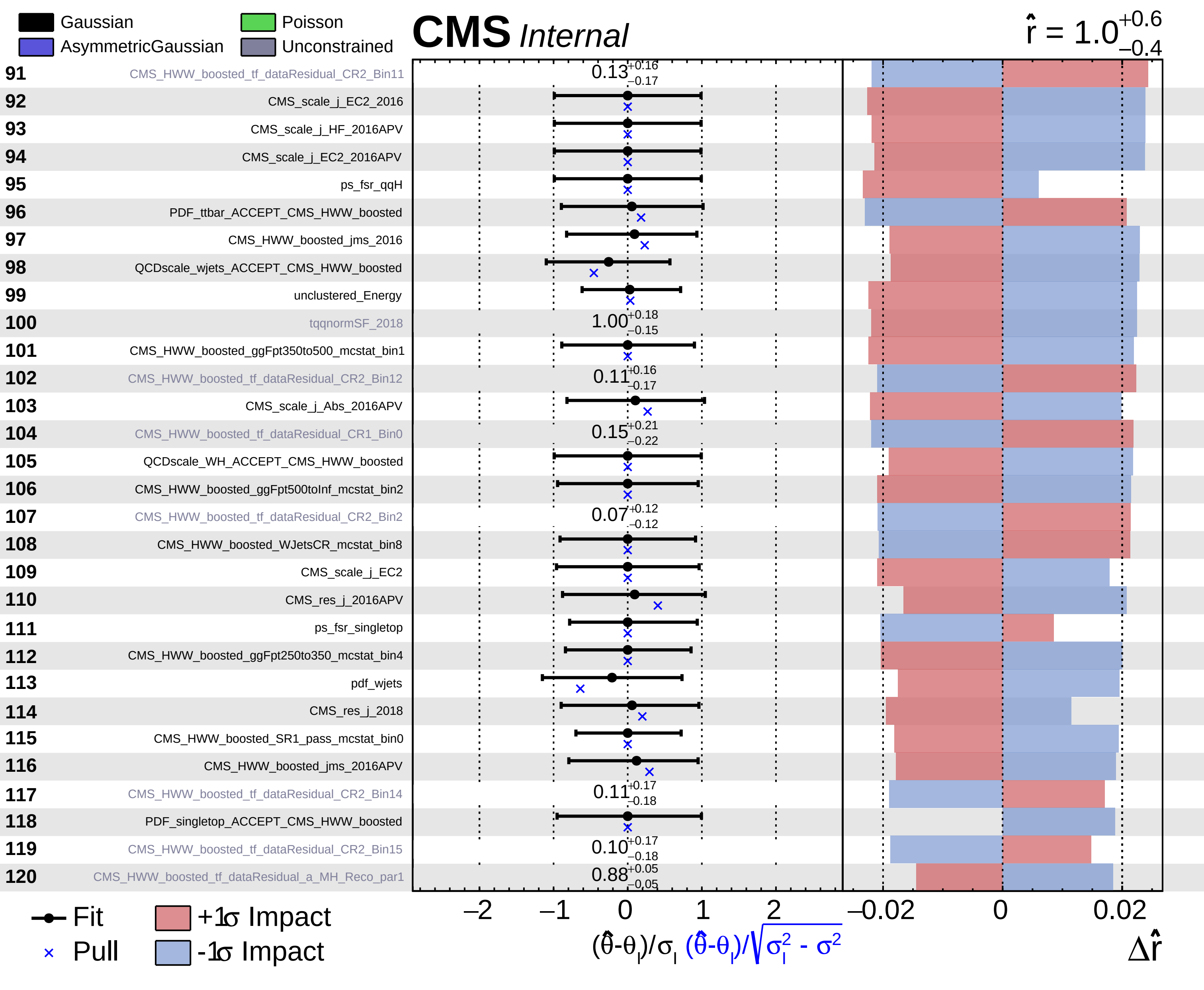

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
1 CMS_HWW_boosted_taggereff
2 CMS_HWW_boosted_SR1_pass_mcstat_bin2
3 CMS_HWW_boosted_VBF_mcstat_bin3
4 ttbarnormSF $1.00^{+0.10}_{-0.09}$
5 CMS_HWW_boosted_TopPtReweight
6 CMS_HWW_boosted_SR1_pass_mcstat_bin1
7 CMS_pileup_2018
8 CMS_HWW_boosted_btagSFbcCorrelated
9 CMS_HWW_boosted_VBF_mcstat_bin2
10 CMS_HWW_boosted_tf_dataResidual_CR1_Bin5 $0.0^{+0.6}_{-0.6}$
11 CMS_HWW_boosted_tf_dataResidual_CR1_Bin6 $0.0^{+0.5}_{-0.5}$
12 CMS_HWW_boosted_tf_dataResidual_CR1_Bin9 $0.00^{+0.31}_{-0.33}$
13 CMS_HWW_boosted_tf_dataResidual_CR1_Bin7 $0.0^{+0.4}_{-0.4}$
14 CMS_HWW_boosted_VBF_mcstat_bin4
15 CMS_HWW_boosted_tf_dataResidual_CR1_Bin8 $0.00^{+0.32}_{-0.34}$
16 ps_fsr_wjets
17 CMS_HWW_boosted_tf_dataResidual_CR1_Bin10 $0.27^{+0.30}_{-0.33}$
18 PDF_qqH_ACCEPT_CMS_HWW_boosted
19 CMS_HWW_boosted_tf_dataResidual_CR1_Bin11 $0.29^{+0.31}_{-0.33}$
20 CMS_HWW_boosted_SR1_pass_mcstat_bin3
21 CMS_HWW_boosted_VBF_mcstat_bin1
22 CMS_HWW_boosted_tf_dataResidual_CR1_Bin18 $0.16^{+0.34}_{-0.36}$
23 CMS_HWW_boosted_tf_dataResidual_CR1_Bin4 $0.0^{+0.7}_{-0.7}$
24 CMS_HWW_boosted_tf_dataResidual_CR1_Bin16 $0.2^{+0.4}_{-0.4}$
25 CMS_HWW_boosted_tf_dataResidual_CR1_Bin1 $0.20^{+0.25}_{-0.27}$
26 CMS_HWW_boosted_tf_dataResidual_CR1_Bin19 $0.16^{+0.33}_{-0.35}$
27 CMS_HWW_boosted_btagSFbc_2017
28 CMS_scale_j_RelSample_2018
29 CMS_HWW_boosted_tf_dataResidual_CR1_Bin12 $0.26^{+0.31}_{-0.34}$
30 CMS_HWW_boosted_ggFpt350to500_mcstat_bin3
−2 −1 0 1 2
−0.2 0 0.2
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

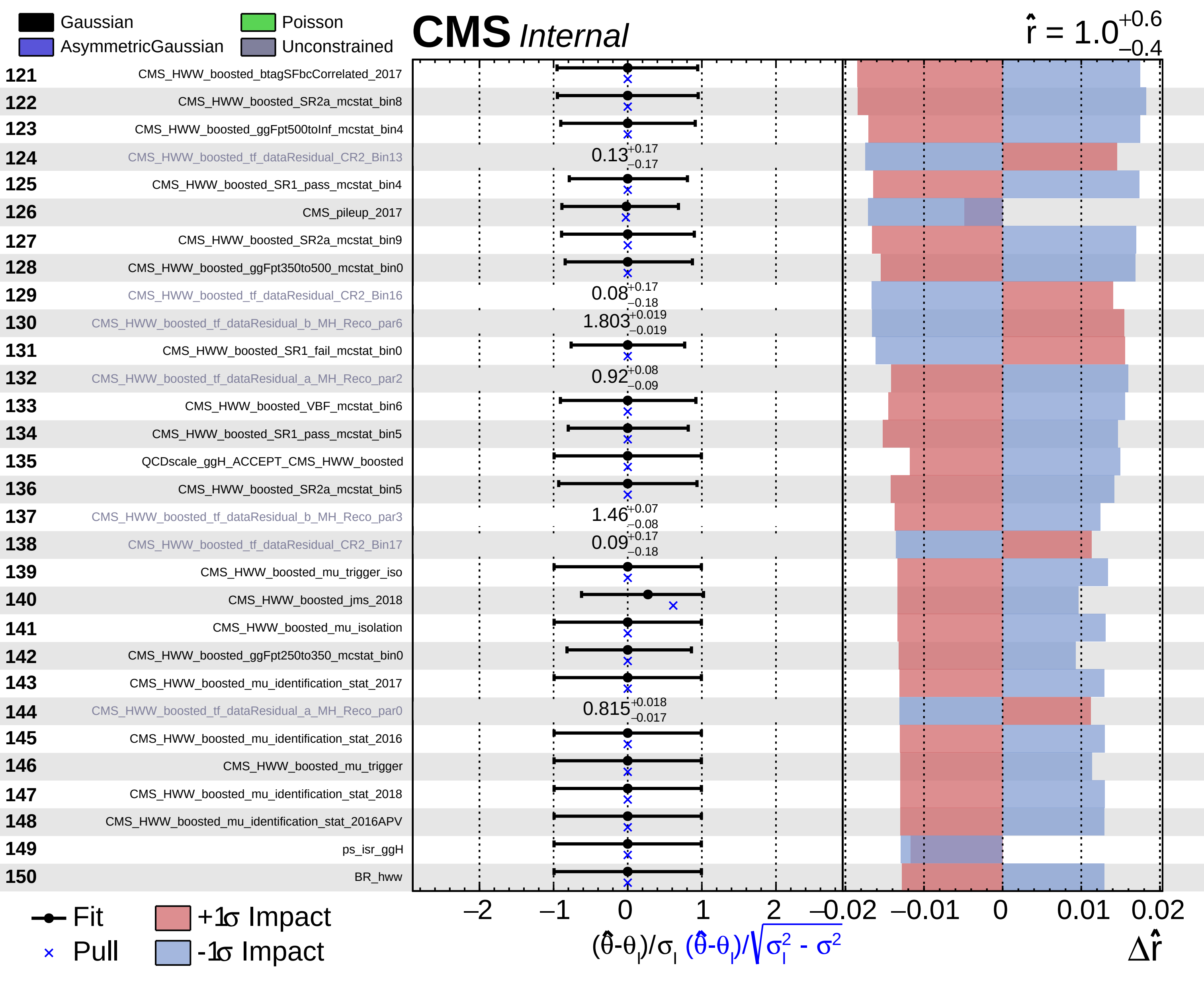

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
31 CMS_res_j_2017
32 CMS_scale_j_BBEC1_2017
33 CMS_HWW_boosted_tf_dataResidual_CR1_Bin15 $0.2^{+0.4}_{-0.4}$
34 CMS_l1_ecal_prefiring_2017
35 CMS_HWW_boosted_tf_dataResidual_CR1_Bin2 $0.25^{+0.30}_{-0.32}$
36 CMS_HWW_boosted_jms_2017
37 CMS_HWW_boosted_btagSFbc_2018
38 CMS_HWW_boosted_tagger
39 CMS_HWW_boosted_WJetsCR_mcstat_bin7
40 CMS_HWW_boosted_WJetsCR_mcstat_bin5
41 CMS_HWW_boosted_ggFpt350to500_mcstat_bin2
42 QCDscale_wjets
43 CMS_HWW_boosted_tf_dataResidual_CR2_Bin6 $0.00^{+0.16}_{-0.17}$
44 CMS_HWW_boosted_tf_dataResidual_CR2_Bin5 $0.00^{+0.16}_{-0.17}$
45 CMS_scale_j_HF_2017
46 CMS_HWW_boosted_tf_dataResidual_CR2_Bin7 $0.00^{+0.16}_{-0.17}$
47 QCDscale_qqH_ACCEPT_CMS_HWW_boosted
48 CMS_scale_j_Abs_2017
49 CMS_HWW_boosted_WJetsCR_mcstat_bin6
50 CMS_scale_j_HF
51 CMS_scale_j_RelSample_2017
52 CMS_scale_j_EC2_2017
53 CMS_HWW_boosted_tf_dataResidual_CR2_Bin4 $0.00^{+0.16}_{-0.17}$
54 CMS_HWW_boosted_tf_dataResidual_CR1_Bin3 $0.5^{+0.5}_{-0.6}$
55 ps_fsr_ggH
56 CMS_HWW_boosted_FakeRate_statistical_uncertainty
57 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par3 $1.77^{+0.05}_{-0.05}$
58 CMS_scale_j_BBEC1_2018
59 CMS_HWW_boosted_jmr_2017
60 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin3
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

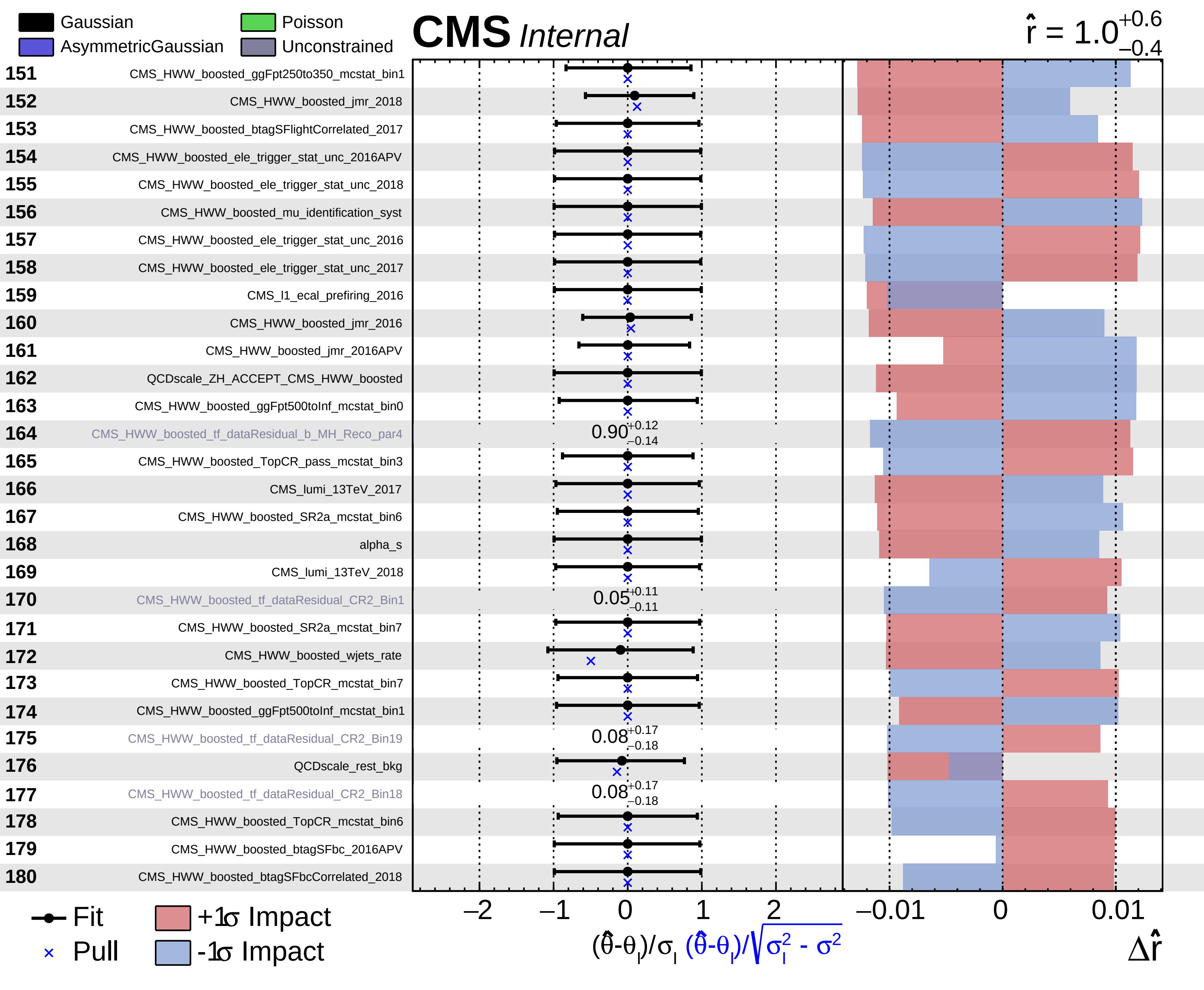


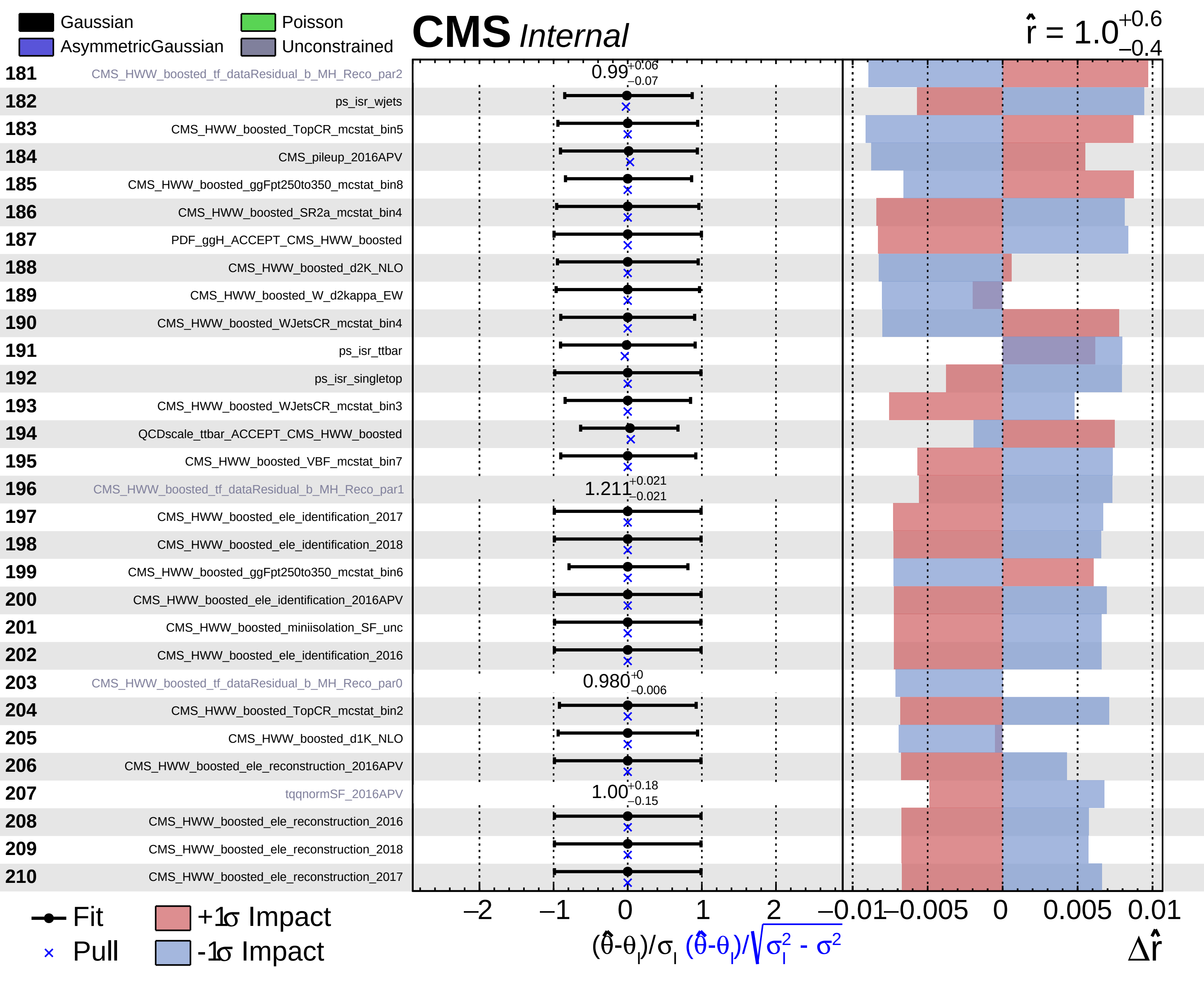
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
61 CMS_HWW_boosted_tf_dataResidual_CR2_Bin8 $0.00^{+0.16}_{-0.17}$
62 CMS_HWW_boosted_tf_dataResidual_CR1_Bin13 $0.26^{+0.34}_{-0.36}$
63 CMS_HWW_boosted_ele_trigger_syst_unc
64 CMS_scale_j_HF_2018
65 CMS_scale_j_RelBal
66 CMS_HWW_boosted_tf_dataResidual_CR1_Bin14 $0.2^{+0.4}_{-0.4}$
67 QCDscale_singletop_ACCEPT_CMS_HWW_boosted
68 CMS_HWW_boosted_tf_dataResidual_CR2_Bin9 $0.00^{+0.16}_{-0.17}$
69 CMS_scale_j_BBEC1
70 CMS_HWW_boosted_ggFpt350to500_mcstat_bin4
71 CMS_scale_j_RelSample_2016APV
72 CMS_scale_j_EC2_2018
73 CMS_scale_j_Abs_2016
74 CMS_HWW_boosted_tf_dataResidual_CR2_Bin3 $0.11^{+0.14}_{-0.15}$
75 CMS_HWW_boosted_lp_sf_region_a
76 CMS_pileup_id
77 wjetsnormSF $1.00^{+0.13}_{-0.12}$
78 CMS_HWW_boosted_tf_dataResidual_CR2_Bin10 $0.13^{+0.16}_{-0.17}$
79 CMS_scale_j_FlavQCD
80 CMS_HWW_boosted_VBF_mcstat_bin5
81 CMS_res_j_2016
82 CMS_scale_j_Abs
83 CMS_HWW_boosted_VBF_mcstat_bin0
84 CMS_HWW_boosted_ggFpt250to350_mcstat_bin2
85 CMS_scale_j_BBEC1_2016APV
86 CMS_scale_j_RelSample_2016
87 CMS_scale_j_HF_2016
88 CMS_scale_j_BBEC1_2016
89 CMS_HWW_boosted_ggFpt250to350_mcstat_bin3
90 QCDscale_ttbar
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



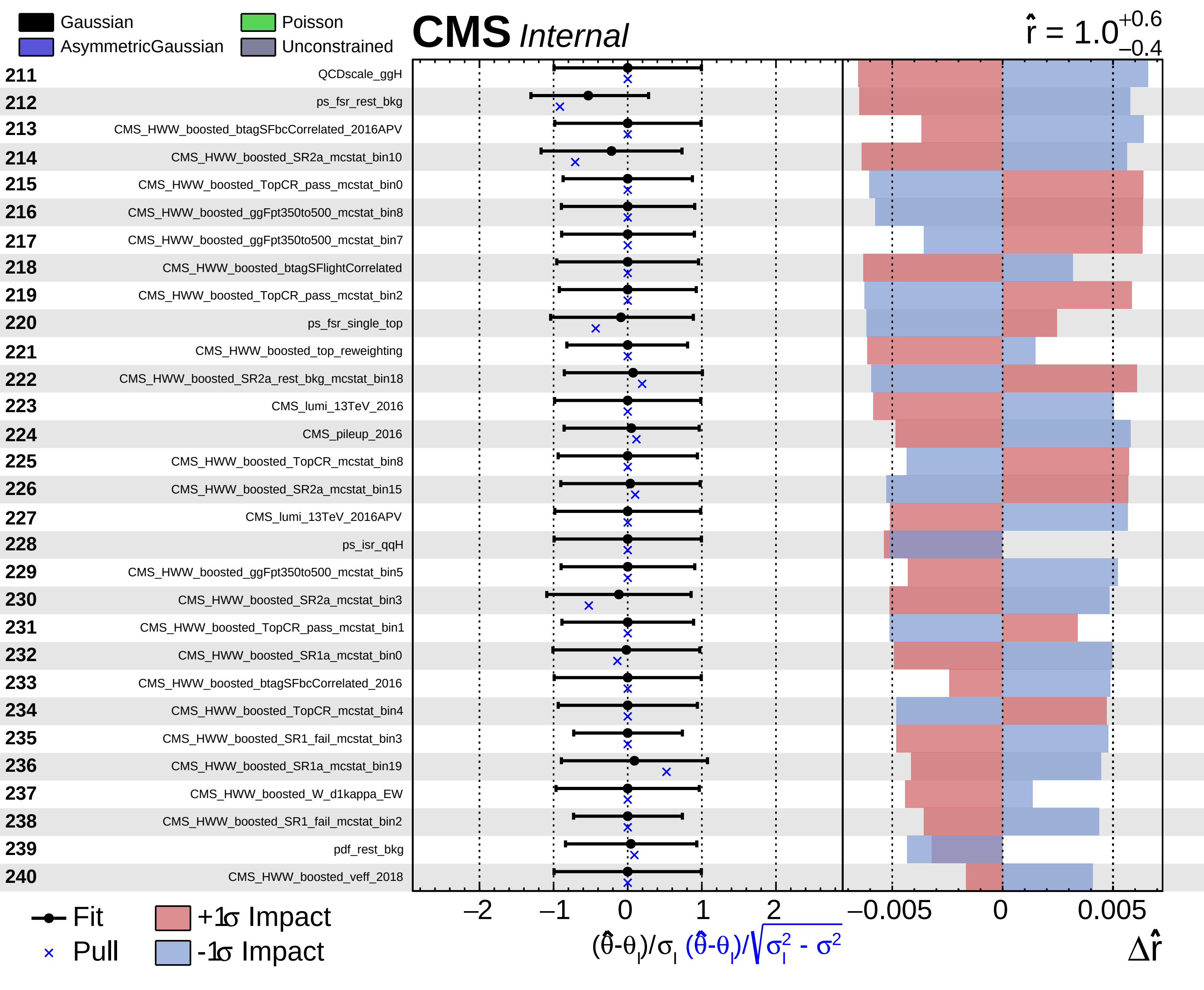
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 1.0+0.6−0.4
91 CMS_HWW_boosted_tf_dataResidual_CR2_Bin11
0.13+0.16−0.17
92 CMS_scale_j_EC2_2016
93 CMS_scale_j_HF_2016APV
94 CMS_scale_j_EC2_2016APV
95 ps_fsr_qqH
96 PDF_ttbar_ACCEPT_CMS_HWW_boosted
97 CMS_HWW_boosted_jms_2016
98 QCDscale_wjets_ACCEPT_CMS_HWW_boosted
99 unclustered_Energy
100 tqqnormSF_2018
1.00+0.18−0.15
101 CMS_HWW_boosted_ggFpt350to500_mcstat_bin1
102 CMS_HWW_boosted_tf_dataResidual_CR2_Bin12
0.11+0.16−0.17
103 CMS_scale_j_Abs_2016APV
104 CMS_HWW_boosted_tf_dataResidual_CR1_Bin0
0.15+0.21−0.22
105 QCDscale_WH_ACCEPT_CMS_HWW_boosted
106 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin2
107 CMS_HWW_boosted_tf_dataResidual_CR2_Bin2
0.07+0.12−0.12
108 CMS_HWW_boosted_WJetsCR_mcstat_bin8
109 CMS_scale_j_EC2
110 CMS_res_j_2016APV
111 ps_fsr_singletop
112 CMS_HWW_boosted_ggFpt250to350_mcstat_bin4
113 pdf_wjets
114 CMS_res_j_2018
115 CMS_HWW_boosted_SR1_pass_mcstat_bin0
116 CMS_HWW_boosted_jms_2016APV
117 CMS_HWW_boosted_tf_dataResidual_CR2_Bin14
0.11+0.17−0.18
118 PDF_singletop_ACCEPT_CMS_HWW_boosted
119 CMS_HWW_boosted_tf_dataResidual_CR2_Bin15
0.10+0.17−0.18
120 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par1
0.88+0.05−0.05
−2
−1
0
1
2
−0.02
0
0.02
Fit
+1σ Impact
Pull
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

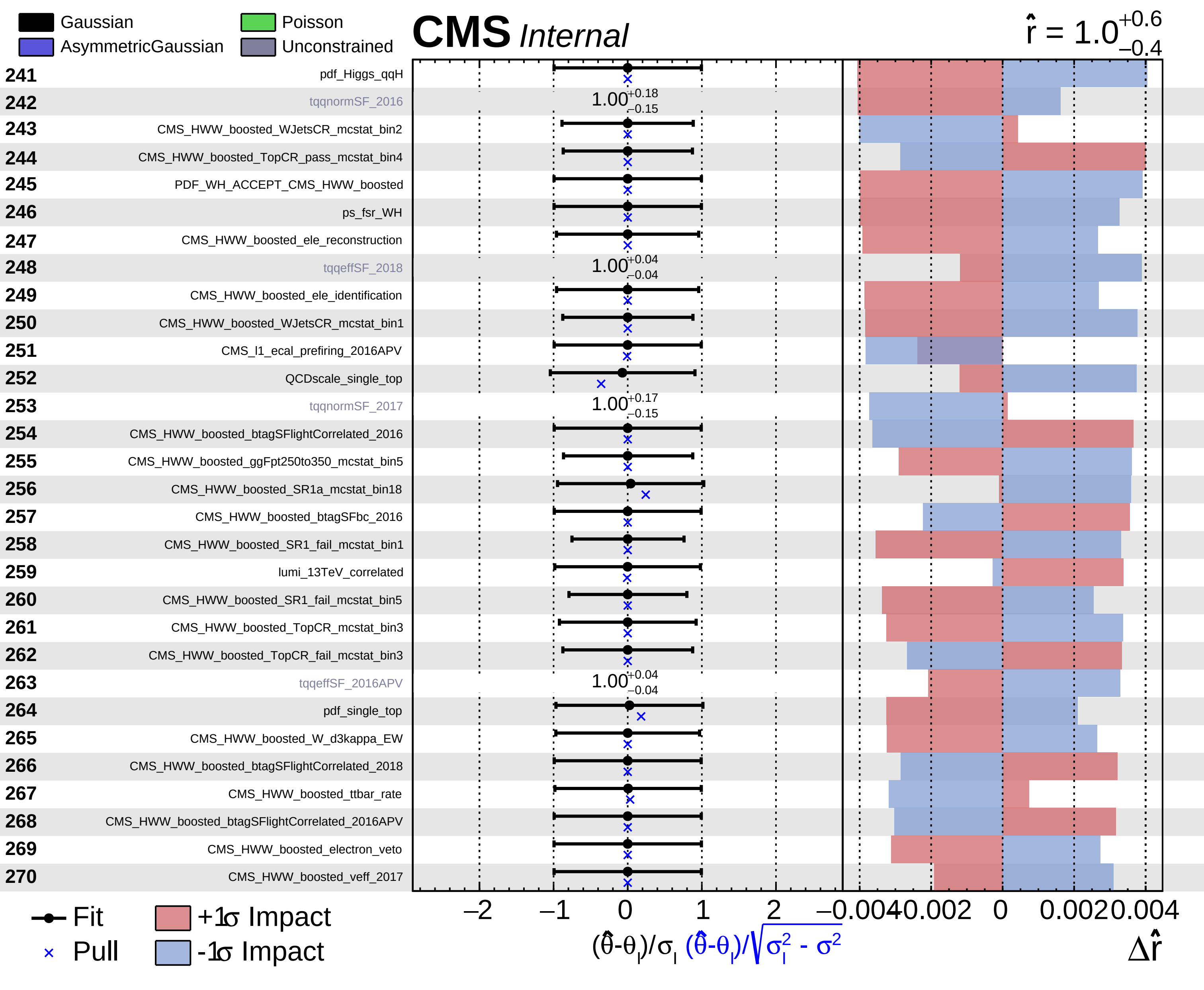

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
121 CMS_HWW_boosted_btagSFbcCorrelated_2017
122 CMS_HWW_boosted_SR2a_mcstat_bin8
123 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin4
124 CMS_HWW_boosted_tf_dataResidual_CR2_Bin13
$0.13^{+0.17}_{-0.17}$
125 CMS_HWW_boosted_SR1_pass_mcstat_bin4
126 CMS_pileup_2017
127 CMS_HWW_boosted_SR2a_mcstat_bin9
128 CMS_HWW_boosted_ggFpt350to500_mcstat_bin0
129 CMS_HWW_boosted_tf_dataResidual_CR2_Bin16
$0.08^{+0.17}_{-0.18}$
130 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par6
$1.803^{+0.019}_{-0.019}$
131 CMS_HWW_boosted_SR1_fail_mcstat_bin0
132 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par2
$0.92^{+0.08}_{-0.09}$
133 CMS_HWW_boosted_VBF_mcstat_bin6
134 CMS_HWW_boosted_SR1_pass_mcstat_bin5
135 QCDscale_ggH_ACCEPT_CMS_HWW_boosted
136 CMS_HWW_boosted_SR2a_mcstat_bin5
137 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par3
$1.46^{+0.07}_{-0.08}$
138 CMS_HWW_boosted_tf_dataResidual_CR2_Bin17
$0.09^{+0.17}_{-0.18}$
139 CMS_HWW_boosted_mu_trigger_iso
140 CMS_HWW_boosted_jms_2018
141 CMS_HWW_boosted_mu_isolation
142 CMS_HWW_boosted_ggFpt250to350_mcstat_bin0
143 CMS_HWW_boosted_mu_identification_stat_2017
144 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par0
$0.815^{+0.018}_{-0.017}$
145 CMS_HWW_boosted_mu_identification_stat_2016
146 CMS_HWW_boosted_mu_trigger
147 CMS_HWW_boosted_mu_identification_stat_2018
148 CMS_HWW_boosted_mu_identification_stat_2016APV
149 ps_isr_ggH
150 BR_hww
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

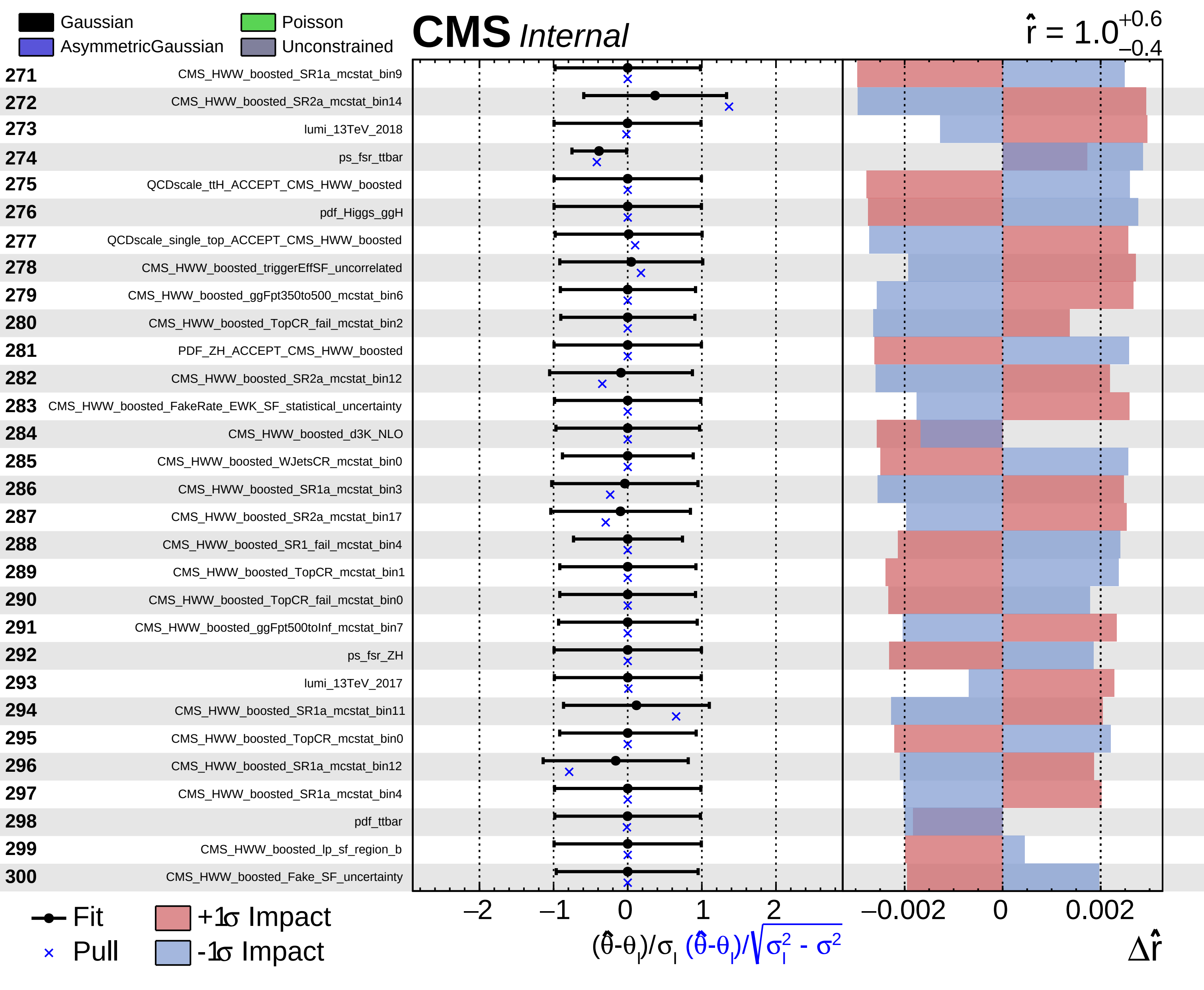

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
151 CMS_HWW_boosted_ggFpt250to350_mcstat_bin1
152 CMS_HWW_boosted_jmr_2018
153 CMS_HWW_boosted_btagSFlightCorrelated_2017
154 CMS_HWW_boosted_ele_trigger_stat_unc_2016APV
155 CMS_HWW_boosted_ele_trigger_stat_unc_2018
156 CMS_HWW_boosted_mu_identification_syst
157 CMS_HWW_boosted_ele_trigger_stat_unc_2016
158 CMS_HWW_boosted_ele_trigger_stat_unc_2017
159 CMS_l1_ecal_prefiring_2016
160 CMS_HWW_boosted_jmr_2016
161 CMS_HWW_boosted_jmr_2016APV
162 QCDscale_ZH_ACCEPT_CMS_HWW_boosted
163 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin0
164 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par4
$0.90^{+0.12}_{-0.14}$
165 CMS_HWW_boosted_TopCR_pass_mcstat_bin3
166 CMS_lumi_13TeV_2017
167 CMS_HWW_boosted_SR2a_mcstat_bin6
168 alpha_s
169 CMS_lumi_13TeV_2018
170 CMS_HWW_boosted_tf_dataResidual_CR2_Bin1
$0.05^{+0.11}_{-0.11}$
171 CMS_HWW_boosted_SR2a_mcstat_bin7
172 CMS_HWW_boosted_wjets_rate
173 CMS_HWW_boosted_TopCR_mcstat_bin7
174 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin1
175 CMS_HWW_boosted_tf_dataResidual_CR2_Bin19
$0.08^{+0.17}_{-0.18}$
176 QCDscale_rest_bkg
177 CMS_HWW_boosted_tf_dataResidual_CR2_Bin18
$0.08^{+0.17}_{-0.18}$
178 CMS_HWW_boosted_TopCR_mcstat_bin6
179 CMS_HWW_boosted_btagSFbc_2016APV
180 CMS_HWW_boosted_btagSFbcCorrelated_2018
−2
−1
0
1
2
−0.01
0
0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



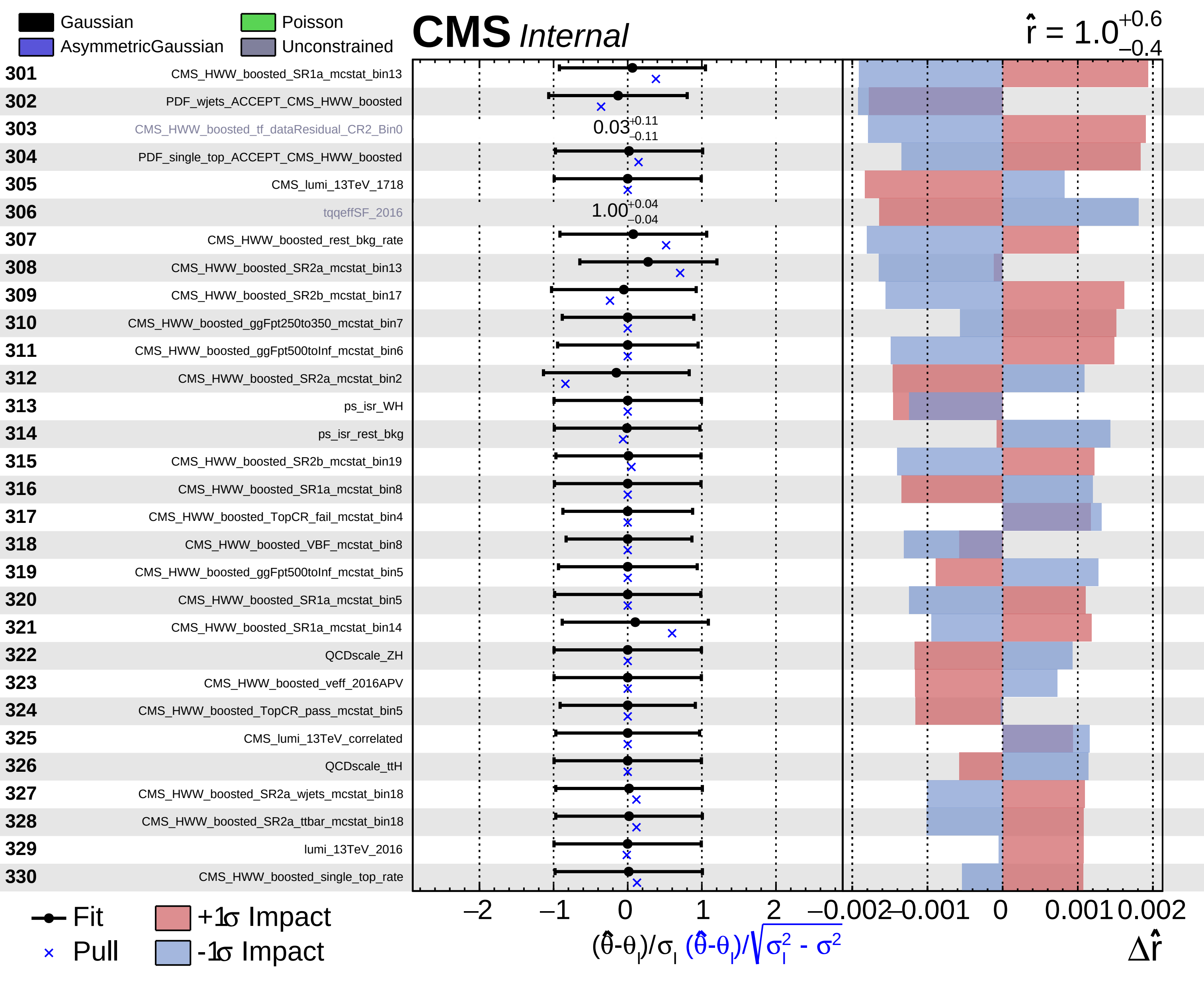
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
181 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par2
$0.99^{+0.06}_{-0.07}$
182 ps_isr_wjets
183 CMS_HWW_boosted_TopCR_mcstat_bin5
184 CMS_pileup_2016APV
185 CMS_HWW_boosted_ggFpt250to350_mcstat_bin8
186 CMS_HWW_boosted_SR2a_mcstat_bin4
187 PDF_ggH_ACCEPT_CMS_HWW_boosted
188 CMS_HWW_boosted_d2K_NLO
189 CMS_HWW_boosted_W_d2kappa_EW
190 CMS_HWW_boosted_WJetsCR_mcstat_bin4
191 ps_isr_ttbar
192 ps_isr_singletop
193 CMS_HWW_boosted_WJetsCR_mcstat_bin3
194 QCDscale_ttbar_ACCEPT_CMS_HWW_boosted
195 CMS_HWW_boosted_VBF_mcstat_bin7
196 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par1
$1.211^{+0.021}_{-0.021}$
197 CMS_HWW_boosted_ele_identification_2017
198 CMS_HWW_boosted_ele_identification_2018
199 CMS_HWW_boosted_ggFpt250to350_mcstat_bin6
200 CMS_HWW_boosted_ele_identification_2016APV
201 CMS_HWW_boosted_miniisolation_SF_unc
202 CMS_HWW_boosted_ele_identification_2016
203 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par0
$0.980^{+0}_{-0.006}$
204 CMS_HWW_boosted_TopCR_mcstat_bin2
205 CMS_HWW_boosted_d1K_NLO
206 CMS_HWW_boosted_ele_reconstruction_2016APV
207 tqqnormSF_2016APV
$1.00^{+0.18}_{-0.15}$
208 CMS_HWW_boosted_ele_reconstruction_2016
209 CMS_HWW_boosted_ele_reconstruction_2018
210 CMS_HWW_boosted_ele_reconstruction_2017
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

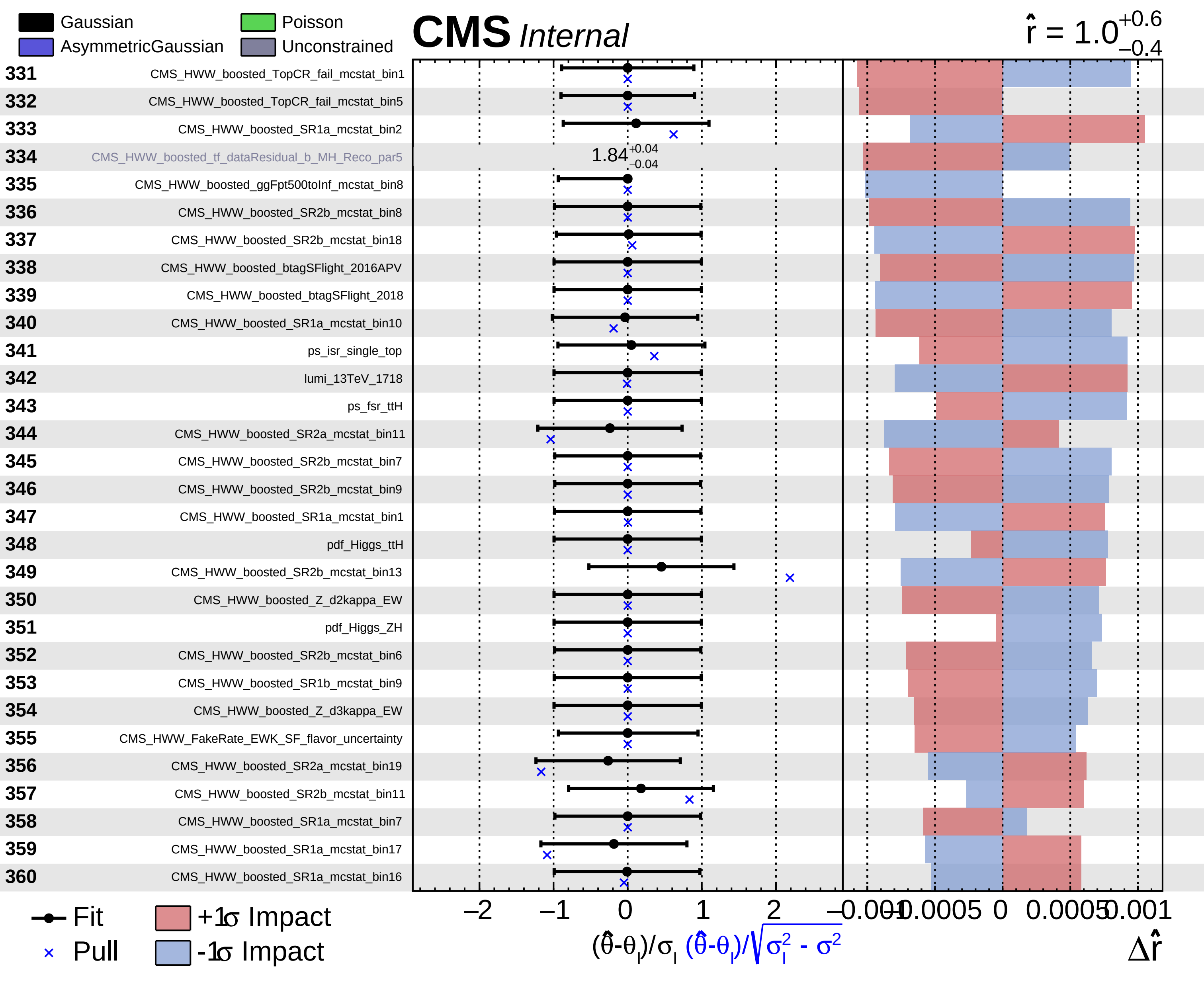

Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
r̂ = 1.0 +0.6 −0.4
211 QCDscale_ggH
212 ps_fsr_rest_bkg
213 CMS_HWW_boosted_btagSFbcCorrelated_2016APV
214 CMS_HWW_boosted_SR2a_mcstat_bin10
215 CMS_HWW_boosted_TopCR_pass_mcstat_bin0
216 CMS_HWW_boosted_ggFpt350to500_mcstat_bin8
217 CMS_HWW_boosted_ggFpt350to500_mcstat_bin7
218 CMS_HWW_boosted_btagSFlightCorrelated
219 CMS_HWW_boosted_TopCR_pass_mcstat_bin2
220 ps_fsr_single_top
221 CMS_HWW_boosted_top_reweighting
222 CMS_HWW_boosted_SR2a_rest_bkg_mcstat_bin18
223 CMS_lumi_13TeV_2016
224 CMS_pileup_2016
225 CMS_HWW_boosted_TopCR_mcstat_bin8
226 CMS_HWW_boosted_SR2a_mcstat_bin15
227 CMS_lumi_13TeV_2016APV
228 ps_isr_qqH
229 CMS_HWW_boosted_ggFpt350to500_mcstat_bin5
230 CMS_HWW_boosted_SR2a_mcstat_bin3
231 CMS_HWW_boosted_TopCR_pass_mcstat_bin1
232 CMS_HWW_boosted_SR1a_mcstat_bin0
233 CMS_HWW_boosted_btagSFbcCorrelated_2016
234 CMS_HWW_boosted_TopCR_mcstat_bin4
235 CMS_HWW_boosted_SR1_fail_mcstat_bin3
236 CMS_HWW_boosted_SR1a_mcstat_bin19
237 CMS_HWW_boosted_W_d1kappa_EW
238 CMS_HWW_boosted_SR1_fail_mcstat_bin2
239 pdf_rest_bkg
240 CMS_HWW_boosted_veff_2018
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

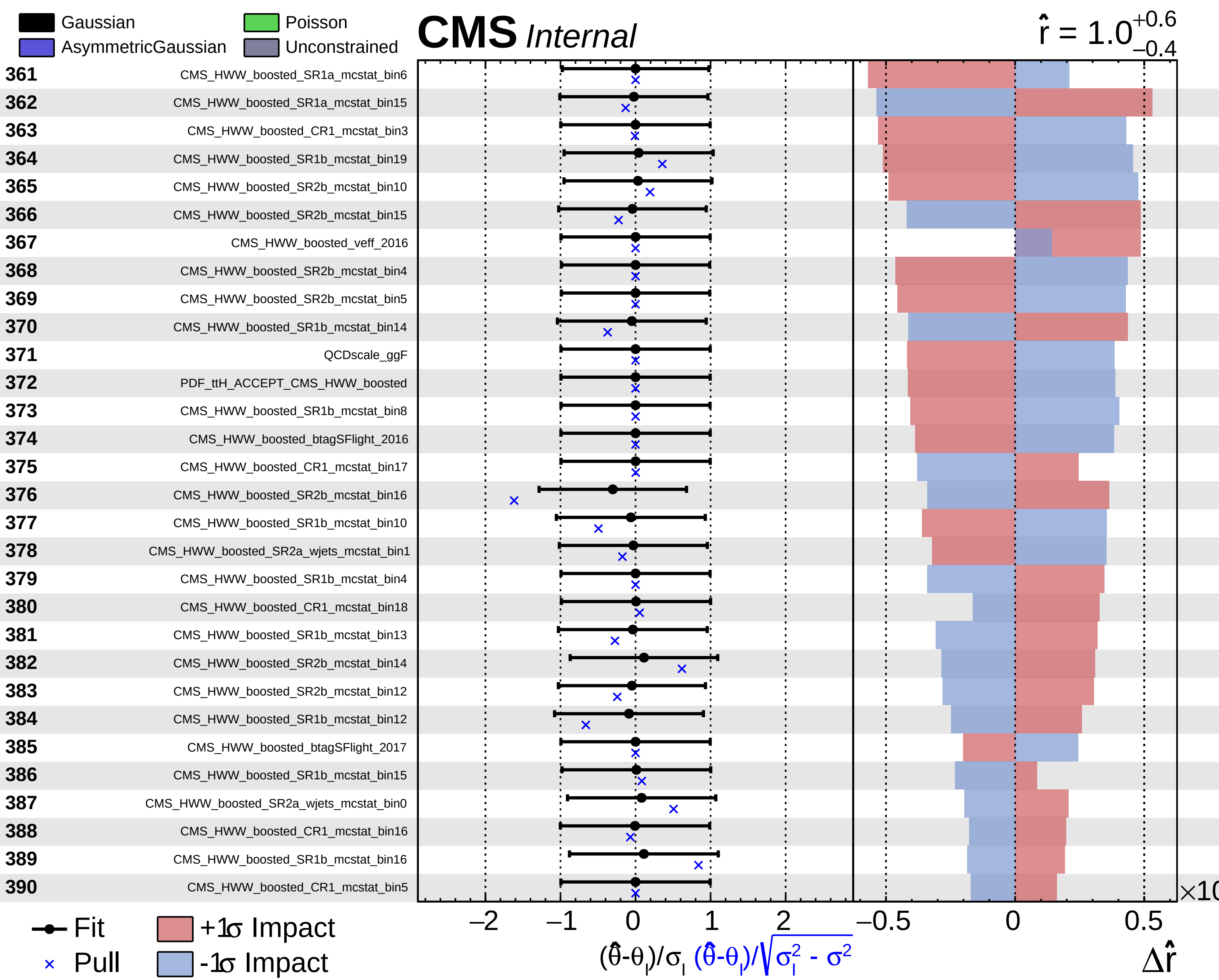

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 1.0+0.6 −0.4
241 pdf_Higgs_qqH
242 tqqnormSF_2016
1.00+0.18 −0.15
243 CMS_HWW_boosted_WJetsCR_mcstat_bin2
244 CMS_HWW_boosted_TopCR_pass_mcstat_bin4
245 PDF_WH_ACCEPT_CMS_HWW_boosted
246 ps_fsr_WH
247 CMS_HWW_boosted_ele_reconstruction
248 tqqeffSF_2018
1.00+0.04 −0.04
249 CMS_HWW_boosted_ele_identification
250 CMS_HWW_boosted_WJetsCR_mcstat_bin1
251 CMS_l1_ecal_prefiring_2016APV
252 QCDscale_single_top
253 tqqnormSF_2017
1.00+0.17 −0.15
254 CMS_HWW_boosted_btagSFlightCorrelated_2016
255 CMS_HWW_boosted_ggFpt250to350_mcstat_bin5
256 CMS_HWW_boosted_SR1a_mcstat_bin18
257 CMS_HWW_boosted_btagSFbc_2016
258 CMS_HWW_boosted_SR1_fail_mcstat_bin1
259 lumi_13TeV_correlated
260 CMS_HWW_boosted_SR1_fail_mcstat_bin5
261 CMS_HWW_boosted_TopCR_mcstat_bin3
262 CMS_HWW_boosted_TopCR_fail_mcstat_bin3
263 tqqeffSF_2016APV
1.00+0.04 −0.04
264 pdf_single_top
265 CMS_HWW_boosted_W_d3kappa_EW
266 CMS_HWW_boosted_btagSFlightCorrelated_2018
267 CMS_HWW_boosted_ttbar_rate
268 CMS_HWW_boosted_btagSFlightCorrelated_2016APV
269 CMS_HWW_boosted_electron_veto
270 CMS_HWW_boosted_veff_2017
−2
−1
0
1
2
−0.004
0.002
0
0.002
0.004
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂



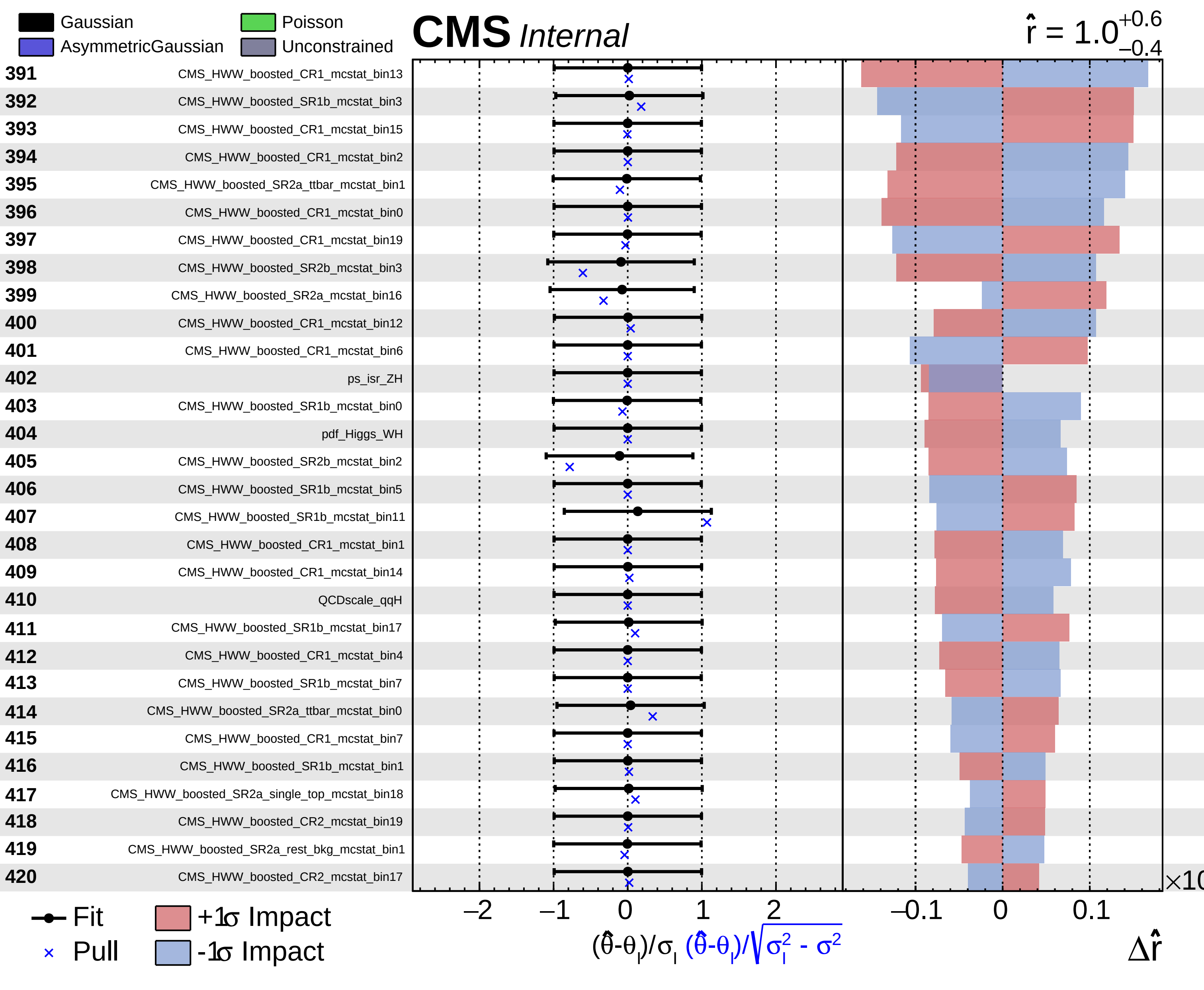
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
271 CMS_HWW_boosted_SR1a_mcstat_bin9
272 CMS_HWW_boosted_SR2a_mcstat_bin14
273 lumi_13TeV_2018
274 ps_fsr_ttbar
275 QCDscale_ttH_ACCEPT_CMS_HWW_boosted
276 pdf_Higgs_ggH
277 QCDscale_single_top_ACCEPT_CMS_HWW_boosted
278 CMS_HWW_boosted_triggerEffSF_uncorrelated
279 CMS_HWW_boosted_ggFpt350to500_mcstat_bin6
280 CMS_HWW_boosted_TopCR_fail_mcstat_bin2
281 PDF_ZH_ACCEPT_CMS_HWW_boosted
282 CMS_HWW_boosted_SR2a_mcstat_bin12
283 CMS_HWW_boosted_FakeRate_EWK_SF_statistical_uncertainty
284 CMS_HWW_boosted_d3K_NLO
285 CMS_HWW_boosted_WJetsCR_mcstat_bin0
286 CMS_HWW_boosted_SR1a_mcstat_bin3
287 CMS_HWW_boosted_SR2a_mcstat_bin17
288 CMS_HWW_boosted_SR1_fail_mcstat_bin4
289 CMS_HWW_boosted_TopCR_mcstat_bin1
290 CMS_HWW_boosted_TopCR_fail_mcstat_bin0
291 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin7
292 ps_fsr_ZH
293 lumi_13TeV_2017
294 CMS_HWW_boosted_SR1a_mcstat_bin11
295 CMS_HWW_boosted_TopCR_mcstat_bin0
296 CMS_HWW_boosted_SR1a_mcstat_bin12
297 CMS_HWW_boosted_SR1a_mcstat_bin4
298 pdf_ttbar
299 CMS_HWW_boosted_lp_sf_region_b
300 CMS_HWW_boosted_Fake_SF_uncertainty
−2 −1 0 1 2
−0.002 0 0.002
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

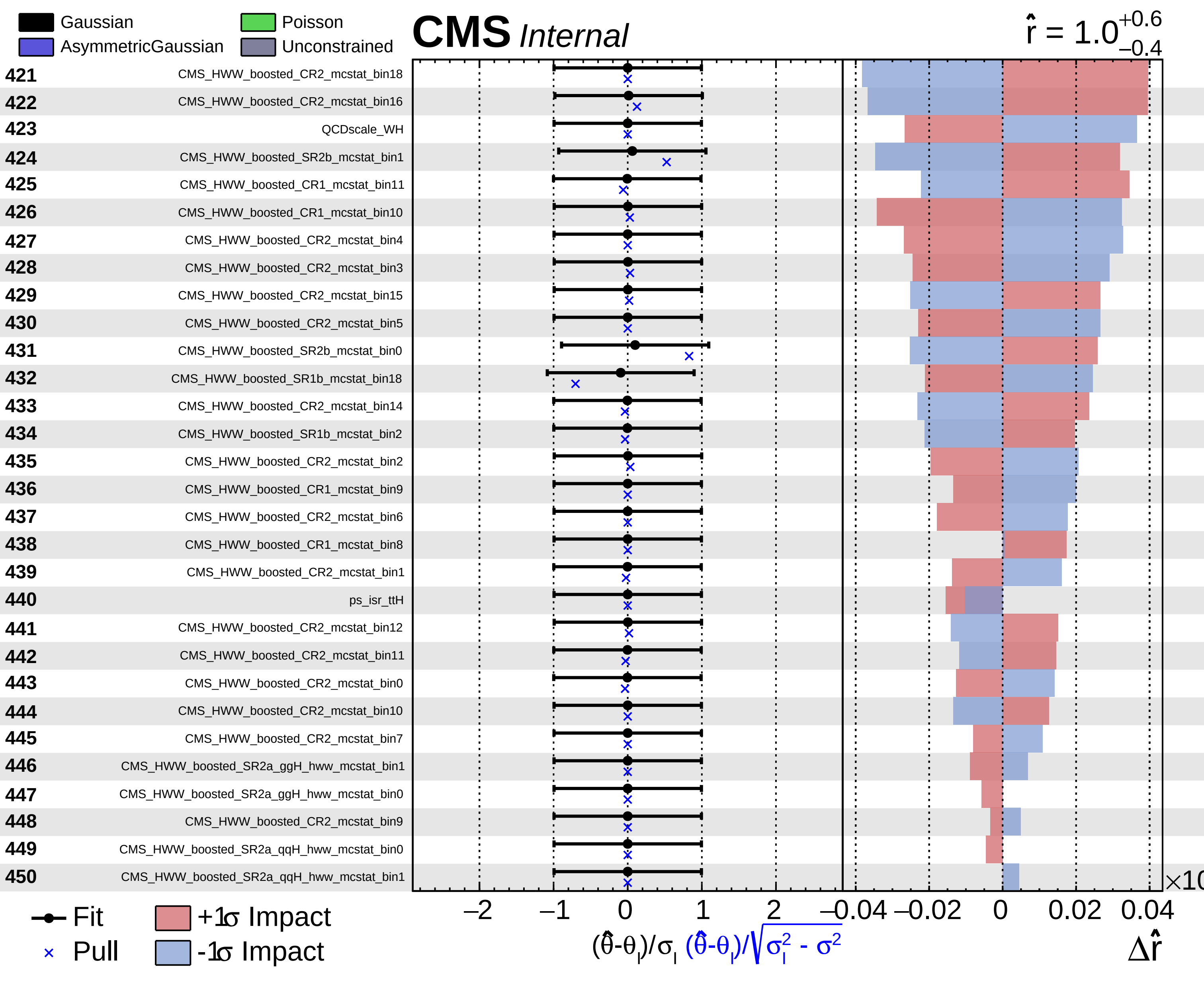

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
301 CMS_HWW_boosted_SR1a_mcstat_bin13
302 PDF_wjets_ACCEPT_CMS_HWW_boosted
303 CMS_HWW_boosted_tf_dataResidual_CR2_Bin0
$0.03^{+0.11}_{-0.11}$
304 PDF_single_top_ACCEPT_CMS_HWW_boosted
305 CMS_lumi_13TeV_1718
306 tqqeffSF_2016
$1.00^{+0.04}_{-0.04}$
307 CMS_HWW_boosted_rest_bkg_rate
308 CMS_HWW_boosted_SR2a_mcstat_bin13
309 CMS_HWW_boosted_SR2b_mcstat_bin17
310 CMS_HWW_boosted_ggFpt250to350_mcstat_bin7
311 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin6
312 CMS_HWW_boosted_SR2a_mcstat_bin2
313 ps_isr_WH
314 ps_isr_rest_bkg
315 CMS_HWW_boosted_SR2b_mcstat_bin19
316 CMS_HWW_boosted_SR1a_mcstat_bin8
317 CMS_HWW_boosted_TopCR_fail_mcstat_bin4
318 CMS_HWW_boosted_VBF_mcstat_bin8
319 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin5
320 CMS_HWW_boosted_SR1a_mcstat_bin5
321 CMS_HWW_boosted_SR1a_mcstat_bin14
322 QCDscale_ZH
323 CMS_HWW_boosted_veff_2016APV
324 CMS_HWW_boosted_TopCR_pass_mcstat_bin5
325 CMS_lumi_13TeV_correlated
326 QCDscale_ttH
327 CMS_HWW_boosted_SR2a_wjets_mcstat_bin18
328 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin18
329 lumi_13TeV_2016
330 CMS_HWW_boosted_single_top_rate
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
331 CMS_HWW_boosted_TopCR_fail_mcstat_bin1
332 CMS_HWW_boosted_TopCR_fail_mcstat_bin5
333 CMS_HWW_boosted_SR1a_mcstat_bin2
334 CMS_HWW_boosted_tf_dataResidual_b_MH_Reco_par5
$1.84^{+0.04}_{-0.04}$
335 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin8
336 CMS_HWW_boosted_SR2b_mcstat_bin8
337 CMS_HWW_boosted_SR2b_mcstat_bin18
338 CMS_HWW_boosted_btagSFlight_2016APV
339 CMS_HWW_boosted_btagSFlight_2018
340 CMS_HWW_boosted_SR1a_mcstat_bin10
341 ps_isr_single_top
342 lumi_13TeV_1718
343 ps_fsr_ttH
344 CMS_HWW_boosted_SR2a_mcstat_bin11
345 CMS_HWW_boosted_SR2b_mcstat_bin7
346 CMS_HWW_boosted_SR2b_mcstat_bin9
347 CMS_HWW_boosted_SR1a_mcstat_bin1
348 pdf_Higgs_ttH
349 CMS_HWW_boosted_SR2b_mcstat_bin13
350 CMS_HWW_boosted_Z_d2kappa_EW
351 pdf_Higgs_ZH
352 CMS_HWW_boosted_SR2b_mcstat_bin6
353 CMS_HWW_boosted_SR1b_mcstat_bin9
354 CMS_HWW_boosted_Z_d3kappa_EW
355 CMS_HWW_FakeRate_EWK_SF_flavor_uncertainty
356 CMS_HWW_boosted_SR2a_mcstat_bin19
357 CMS_HWW_boosted_SR2b_mcstat_bin11
358 CMS_HWW_boosted_SR1a_mcstat_bin7
359 CMS_HWW_boosted_SR1a_mcstat_bin17
360 CMS_HWW_boosted_SR1a_mcstat_bin16
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

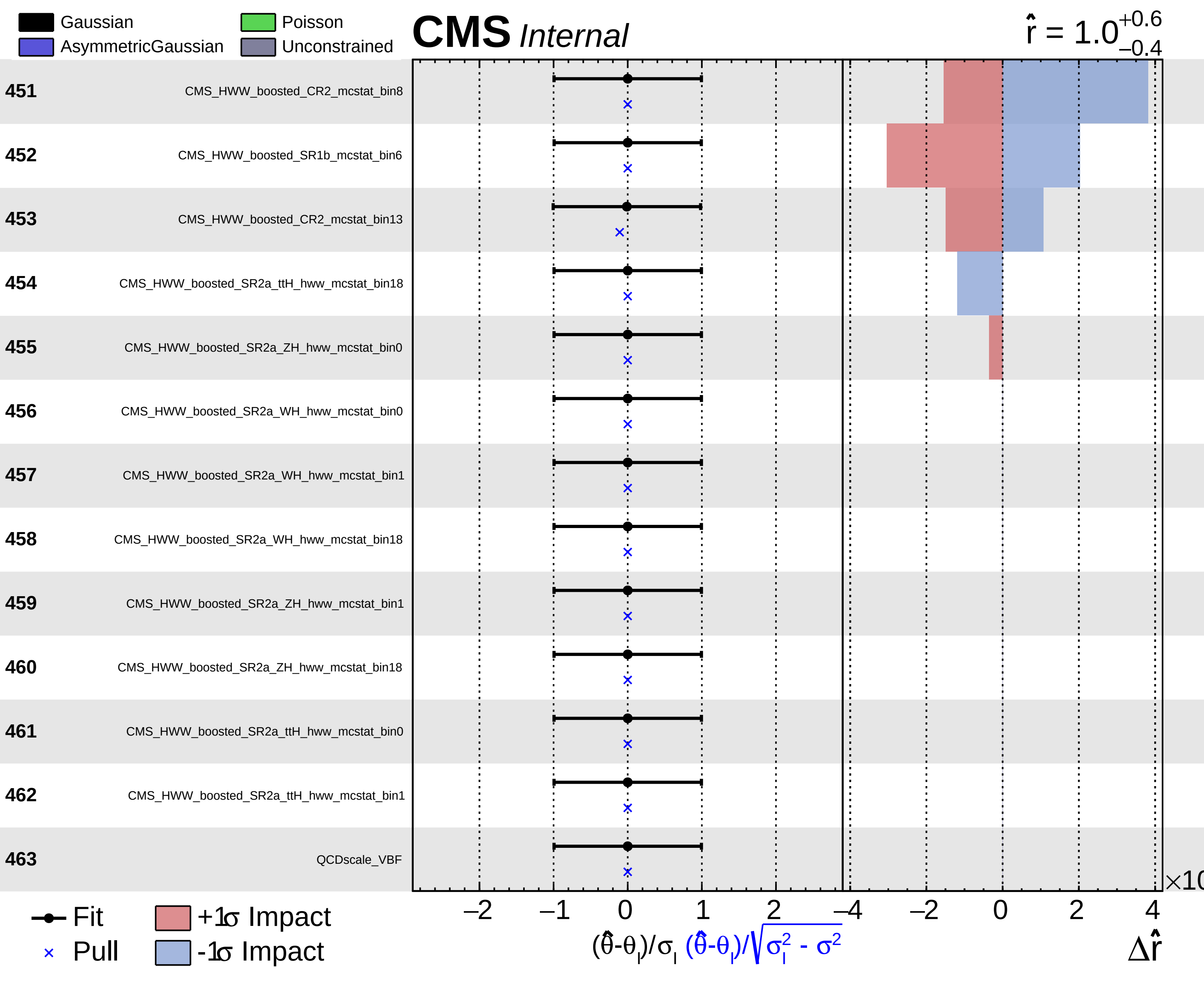


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
361 CMS_HWW_boosted_SR1a_mcstat_bin6
362 CMS_HWW_boosted_SR1a_mcstat_bin15
363 CMS_HWW_boosted_CR1_mcstat_bin3
364 CMS_HWW_boosted_SR1b_mcstat_bin19
365 CMS_HWW_boosted_SR2b_mcstat_bin10
366 CMS_HWW_boosted_SR2b_mcstat_bin15
367 CMS_HWW_boosted_veff_2016
368 CMS_HWW_boosted_SR2b_mcstat_bin4
369 CMS_HWW_boosted_SR2b_mcstat_bin5
370 CMS_HWW_boosted_SR1b_mcstat_bin14
371 QCDscale_ggF
372 PDF_ttH_ACCEPT_CMS_HWW_boosted
373 CMS_HWW_boosted_SR1b_mcstat_bin8
374 CMS_HWW_boosted_btagSFlight_2016
375 CMS_HWW_boosted_CR1_mcstat_bin17
376 CMS_HWW_boosted_SR2b_mcstat_bin16
377 CMS_HWW_boosted_SR1b_mcstat_bin10
378 CMS_HWW_boosted_SR2a_wjets_mcstat_bin1
379 CMS_HWW_boosted_SR1b_mcstat_bin4
380 CMS_HWW_boosted_CR1_mcstat_bin18
381 CMS_HWW_boosted_SR1b_mcstat_bin13
382 CMS_HWW_boosted_SR2b_mcstat_bin14
383 CMS_HWW_boosted_SR2b_mcstat_bin12
384 CMS_HWW_boosted_SR1b_mcstat_bin12
385 CMS_HWW_boosted_btagSFlight_2017
386 CMS_HWW_boosted_SR1b_mcstat_bin15
387 CMS_HWW_boosted_SR2a_wjets_mcstat_bin0
388 CMS_HWW_boosted_CR1_mcstat_bin16
389 CMS_HWW_boosted_SR1b_mcstat_bin16
390 CMS_HWW_boosted_CR1_mcstat_bin5
−2 −1 0 1 2
−0.5 0 0.5
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
391 CMS_HWW_boosted_CR1_mcstat_bin13
392 CMS_HWW_boosted_SR1b_mcstat_bin3
393 CMS_HWW_boosted_CR1_mcstat_bin15
394 CMS_HWW_boosted_CR1_mcstat_bin2
395 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin1
396 CMS_HWW_boosted_CR1_mcstat_bin0
397 CMS_HWW_boosted_CR1_mcstat_bin19
398 CMS_HWW_boosted_SR2b_mcstat_bin3
399 CMS_HWW_boosted_SR2a_mcstat_bin16
400 CMS_HWW_boosted_CR1_mcstat_bin12
401 CMS_HWW_boosted_CR1_mcstat_bin6
402 ps_isr_ZH
403 CMS_HWW_boosted_SR1b_mcstat_bin0
404 pdf_Higgs_WH
405 CMS_HWW_boosted_SR2b_mcstat_bin2
406 CMS_HWW_boosted_SR1b_mcstat_bin5
407 CMS_HWW_boosted_SR1b_mcstat_bin11
408 CMS_HWW_boosted_CR1_mcstat_bin1
409 CMS_HWW_boosted_CR1_mcstat_bin14
410 QCDscale_qqH
411 CMS_HWW_boosted_SR1b_mcstat_bin17
412 CMS_HWW_boosted_CR1_mcstat_bin4
413 CMS_HWW_boosted_SR1b_mcstat_bin7
414 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin0
415 CMS_HWW_boosted_CR1_mcstat_bin7
416 CMS_HWW_boosted_SR1b_mcstat_bin1
417 CMS_HWW_boosted_SR2a_single_top_mcstat_bin18
418 CMS_HWW_boosted_CR2_mcstat_bin19
419 CMS_HWW_boosted_SR2a_rest_bkg_mcstat_bin1
420 CMS_HWW_boosted_CR2_mcstat_bin17
−2
−1
0
1
2
−0.1
0
0.1
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
421 CMS_HWW_boosted_CR2_mcstat_bin18
422 CMS_HWW_boosted_CR2_mcstat_bin16
423 QCDscale_WH
424 CMS_HWW_boosted_SR2b_mcstat_bin1
425 CMS_HWW_boosted_CR1_mcstat_bin11
426 CMS_HWW_boosted_CR1_mcstat_bin10
427 CMS_HWW_boosted_CR2_mcstat_bin4
428 CMS_HWW_boosted_CR2_mcstat_bin3
429 CMS_HWW_boosted_CR2_mcstat_bin15
430 CMS_HWW_boosted_CR2_mcstat_bin5
431 CMS_HWW_boosted_SR2b_mcstat_bin0
432 CMS_HWW_boosted_SR1b_mcstat_bin18
433 CMS_HWW_boosted_CR2_mcstat_bin14
434 CMS_HWW_boosted_SR1b_mcstat_bin2
435 CMS_HWW_boosted_CR2_mcstat_bin2
436 CMS_HWW_boosted_CR1_mcstat_bin9
437 CMS_HWW_boosted_CR2_mcstat_bin6
438 CMS_HWW_boosted_CR1_mcstat_bin8
439 CMS_HWW_boosted_CR2_mcstat_bin1
440 ps_isr_ttH
441 CMS_HWW_boosted_CR2_mcstat_bin12
442 CMS_HWW_boosted_CR2_mcstat_bin11
443 CMS_HWW_boosted_CR2_mcstat_bin0
444 CMS_HWW_boosted_CR2_mcstat_bin10
445 CMS_HWW_boosted_CR2_mcstat_bin7
446 CMS_HWW_boosted_SR2a_ggH_hww_mcstat_bin1
447 CMS_HWW_boosted_SR2a_ggH_hww_mcstat_bin0
448 CMS_HWW_boosted_CR2_mcstat_bin9
449 CMS_HWW_boosted_SR2a_qqH_hww_mcstat_bin0
450 CMS_HWW_boosted_SR2a_qqH_hww_mcstat_bin1
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.4}$
451 CMS_HWW_boosted_CR2_mcstat_bin8
452 CMS_HWW_boosted_SR1b_mcstat_bin6
453 CMS_HWW_boosted_CR2_mcstat_bin13
454 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin18
455 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin0
456 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin0
457 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin1
458 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin18
459 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin1
460 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin18
461 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin0
462 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin1
463 QCDscale_VBF
−2 −1 0 1 2
−4 −2 0 2 4
×10
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$